Md Erfan

PhD Student and Research Assistant, Department of Computer Science, The University of Alabama

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Research Interests

• **Cyber Security:** MITRE ATT&CK, Software Vulnerabilities, Developer Perspectives on Security, Software Supply Chain Security, Cyber Threat Intelligence, and Vulnerability Analysis

• **Software Engineering:** Flaky Test Detection and Mitigation, Test Smells, Software Testing, Compilation Error Resolution, Program Analysis, and Mining Software Repositories

• **Formal Verification and LLMs for Code:** Natural-Language-to-Formally-Verified Code Generation using Large Language Models (LLMs), Formal Specification, Program Verification, and AI-Assisted Verified Software Development

Education

2025–Present **Doctor of Philosophy (PhD) in Computer Science**, *Department of Computer Science, The University of Alabama*, Tuscaloosa, Alabama, USA
Advisor: Dr. Md Rayhanur Rahman

2016 **Master of Science in Software Engineering (MSSE)**, *Institute of Information Technology, University of Dhaka*, Dhaka, Bangladesh, *CGPA: 3.81/4.00; WES Equivalent: 3.97/4.00*
Thesis: *Efficient Runtime Code Offloading Mechanism for Mobile Cloud Computing*
Supervisor: Dr. Md Shariful Islam

2014 **Bachelor of Science in Software Engineering (BSSE)**, *Institute of Information Technology, University of Dhaka*, Dhaka, Bangladesh, *CGPA: 3.80/4.00; WES Equivalent: 3.88/4.00*
Thesis: *Approximating Social Ties Based on Call Logs: Whom Should We Prioritize?*
Supervisor: Dr. Kazi Muhaimin-us-Sakib

Academic and Research Appointments

Fall 2026–Present **Research Assistant**, *Department of Computer Science, The University of Alabama*, Tuscaloosa, Alabama, USA
Conducting research in software engineering, cybersecurity, formal verification, large language models for code, natural-language-to-formally-verified-code generation, software testing, and vulnerability analysis.

Spring 2026 **Teaching Assistant**, *Department of Computer Science, The University of Alabama*, Tuscaloosa, Alabama, USA
Teaching Assistant for Program Analysis and Software Interface Design. Assisted with course instruction, grading, assignments, student questions, and related instructional activities.

Fall 2025 **Teaching Assistant**, *Department of Computer Science, The University of Alabama*, Tuscaloosa, Alabama, USA
Teaching Assistant for Database Systems. Assisted with course instruction, grading, assignments, student questions, and related instructional activities.

Professional Experience

Dec. 2020–Present **Assistant Professor**, *Department of Computer Science and Engineering, University of Barishal*, Bangladesh
Taught undergraduate and graduate courses and supervised student research and projects.

March 2022–March 2025 **Department Head**, *Department of Computer Science and Engineering, University of Barishal*, Bangladesh
Oversaw departmental operations, coordinated faculty activities, and managed academic programs.

2024–July 2025 **Project Coordinator**, *Enhancing Digital Government and Economy (EDGE) Project*, Bangladesh
Managed a project with approximately BDT 5 crore in funding and coordinated faculty members, students, and external stakeholders.

2016–2020 **Lecturer**, *Department of Computer Science and Engineering, University of Barishal*, Bangladesh
Taught undergraduate courses and supervised student research and projects.

2017–2020 **Student Advisor**, *Department of Computer Science and Engineering, University of Barishal*, Bangladesh
Advised undergraduate and graduate students on academic progress, research, and career development.

2015–2016 **Trainer**, *Institute of Information Technology, University of Dhaka*, Dhaka, Bangladesh
Conducted training on web technologies and programming.

2013–2014 **Software Engineer Intern**, *Southtech Limited*, Dhaka, Bangladesh
Developed testing tools and worked on Microsoft SharePoint development.

Research Experience

Summer 2024 **Research Student**, *UIUC+/ASSIP Summer Research Program*

- Conducted research in collaboration with Dr. Wing Lam (George Mason University) and Dr. August Shi (The University of Texas at Austin).
- Automated the end-to-end reproduction of flaky test methods using issue data, code compilation, test execution, dependency analysis, logging, and Dockerized environments for reproducibility.

Publications

Journal Articles

1. **Md Erfan**, Ahmed Ryan, Laurie Williams, and Md Rayhanur Rahman. "Investigating Co-occurrences of MITRE ATT&CK Techniques." *Computers & Security*, 105042, 2026.
2. S. J. Meem, M. Samsuddoha, S. J. Momo, **Md Erfan**, and M. M. Ahmed. "Authorship-Aware Fake News Detection Using Hierarchical Meta-Learning with Transformer-Based Text Features." *Applied Soft Computing*, 116261, 2026.
3. N. Haque, T. Islam, and **Md Erfan**. "An Exploration of Machine Learning Approaches for Early Autism Spectrum Disorder Detection." *Healthcare Analytics*, vol. 7, 100379, 2025.
4. D. M. Setu, T. Islam, **Md Erfan**, S. K. Dey, M. R. A. Asif, and M. Samsuddoha. "A Comprehensive Strategy for Identifying Plagiarism in Academic Submissions." *Journal of Umm Al-Qura University for Engineering and Architecture*, pp. 1–16, 2025.
5. S. Sifath, T. Islam, **Md Erfan**, S. K. Dey, M. M. U. Islam, M. Samsuddoha, and T. Rahman. "Recurrent Neural Network Based Multiclass Cyber Bullying Classification." *Natural Language Processing Journal*, vol. 9, 100111, 2024.
6. A. Islam, M. Samsuddoha, and **Md Erfan**. "WVGM: Water View Google Map, Introducing Water Paths on Rivers to Reach One's Destination Using Various Types of Vehicles." *International Journal of Computer Applications*, vol. 184, no. 45, pp. 8–12, 2023.
7. **Md Erfan**, B. Halder, S. R. Pal, M. S. Islam, and R. H. Faisal. "Experimental Study of Four Selective Code Smells Declining in Real Life Projects." *International Journal of Information and Electronics Engineering*, vol. 8, no. 4, 2018.
8. **Md Erfan**, F. Huq, and M. S. Islam. "An Optimal Task Scheduling Mechanism for Mobile Cloud Computing." *International Journal of Computer Applications*, vol. 175, no. 6, pp. 1–8, 2017.
9. B. A. Hridita, M. Irfan, and M. S. Islam. "Mobility Aware Task Allocation for Mobile Cloud Computing." *International Journal of Computer Applications*, vol. 137, no. 9, pp. 35–41, 2016.

Book Chapters

10. M. W. Haque, **Md Erfan**, S. Dasgupta, M. R. Rahman, and M. Rahman. "Security Vulnerabilities in Software Supply Chain for Autonomous Vehicles." In *Advances in Transportation Cybersecurity and Resiliency*, pp. 97–146, forthcoming 2027.

11. M. Samsuddoha, D. Biswas, and **Md Erfan**. "User Similarity Computation Strategy for Collaborative Filtering Using Word Sense Disambiguation Technique." In *The Fourth Industrial Revolution and Beyond: Select Proceedings of IC4IR+*, pp. 87–101, Springer Nature Singapore, 2023.
12. D. Biswas, M. Samsuddoha, **Md Erfan**, and R. H. Faisal. "Autism Spectrum Disorder Detecting Mechanism on Social Communication Skills Using Machine Learning Approaches." In *Proceedings of SAI Intelligent Systems Conference*, pp. 591–609, Springer Nature Switzerland, 2023.

Conference Publications

13. S. H. Rumman, M. M. Rahman, T. Islam, M. Z. Akon, J. A. Joy, and **Md Erfan**. "Improving Accuracy in Medical Imaging: A YOLO-NAS and SAM Hybrid Approach for Enhanced Medical Image Segmentation." In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pp. 1708–1713, IEEE, 2024.
14. J. A. Joy, Z. Akon, P. S. Sarker, M. M. H. Shaon, M. A. Razzak, et al., **Md Erfan**. "Leveraging Large Language Models for Text-to-SQL with Attention to Data Security." In *International Conference on Machine Intelligence and Emerging Technologies (MIET)*, 2024.
15. M. A. R. Shamim, A. B. M. Anas, and **Md Erfan**. "Identification of Vector and Non-vector Mosquito Species Using Deep Convolutional Neural Networks with Ensemble Model." In *2022 International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, pp. 1–6, IEEE, 2022.
16. B. A. Hridita, **Md Erfan**, and M. S. Islam. "Task Allocation for Mobile Cloud Computing: State-of-the-Art and Open Challenges." In *2016 5th International Conference on Informatics, Electronics and Vision (ICIEV)*, pp. 752–757, IEEE, 2016.
17. **Md Erfan**, A. U. Gias, S. M. Sarwar, and K. Sakib. "Approximating Social Ties Based on Call Logs: Whom Should We Prioritize?" In *AIST (Supplement)*, pp. 28–39, 2015.

Other Publications

18. C. Wang, **Md Erfan**, M. M. H. Pious, S. Rafi, A. Shi, and W. Lam. "Constructing a Comprehensive Dataset of Reproducible Flaky Tests." *Journal of Student-Scientists' Research*, vol. 6, 2024.
19. **Md Erfan**, N. Nahar, B. A. Hridita, S. M. Khaled, and M. S. Islam. "Dynamic Method Level Code Offloading for Performance Improvement and Energy Saving." *Advances in Computing, Control and Communication Technology*, vol. 1, p. 89, 2016.

Preprints and Manuscripts

20. A. Ryan, **Md Erfan**, A. A. U. Rahman, and M. R. Rahman. "Can Open-Weight LLMs Produce Kernel-Verified Coq Proofs? A Pilot Study." *arXiv preprint arXiv:2608.05420*, 2026.
21. **Md Erfan**, A. Ryan, and M. R. Rahman. "A Large-Scale Measurement of AI Bill of Materials Completeness in Hugging Face Models." *arXiv preprint arXiv:2607.17242*, 2026.
22. **Md Erfan**, A. Ryan, M. K. H. Chowdhury, and M. R. Rahman. "Evaluating Open-Weight LLMs for Generating Structured Threat Information for Autonomous Vehicle Vulnerabilities." *arXiv preprint arXiv:2607.16175*, 2026.
23. A. Ryan, S. S. Noor, **Md Erfan**, S. Mitra, S. Mittal, and M. R. Rahman. "Evaluating Open-Source LLMs for Multi-Label ATT&CK Technique Classification on CTI Reports." *arXiv preprint arXiv:2606.18166*, 2026.
24. A. A. Niloy, A. Ryan, I. H. Rafi, **Md Erfan**, and M. R. Rahman. "Multi-Source Cybersecurity Logs: An ATT&CK-Labeled Dataset and SLM Evaluation." *arXiv preprint arXiv:2606.18190*, 2026.
25. S. Rafi, M. U. H. Sumon, **Md Erfan**, M. M. Khan, A. Shi, and W. Lam. "A Dataset of Reproducible Flaky-Test Failures." *arXiv preprint arXiv:2605.21677*, 2026.
26. **Md Erfan**, M. K. H. Chowdhury, A. Ryan, and M. R. Rahman. "From Natural Language to Verified Code: Toward AI-Assisted Problem-to-Code Generation with Dafny-Based Formal Verification." *arXiv preprint arXiv:2604.22601*, 2026.

27. E. Curl, K. Ampomah, **Md Erfan**, and S. Dibbo. "Beyond Attack Success Rate: A Multi-Metric Evaluation of Adversarial Transferability in Medical Imaging Models." *arXiv preprint arXiv:2604.16532*, 2026.
28. M. Tamanna, S. Mitra, **Md Erfan**, A. Ryan, S. Mittal, L. Williams, and M. R. Rahman. "What Are Adversaries Doing? Automating Tactics, Techniques, and Procedures Extraction: A Systematic Review." *arXiv preprint arXiv:2604.02377*, 2026.
29. A. Ryan, I. Khalil, A. Al Jahid, **Md Erfan**, S. Park, A. A. U. Rahman, and M. R. Rahman. "Mind the Gap: Evaluating LLMs for High-Level Malicious Package Detection vs. Fine-Grained Indicator Identification." *arXiv preprint arXiv:2602.16304*, 2026.
30. A. Ryan, J. M. Ifti, **Md Erfan**, A. A. U. Rahman, and M. R. Rahman. "Unveiling Malicious Logic: Towards a Statement-Level Taxonomy and Dataset for Securing Python Packages." *arXiv preprint arXiv:2512.12559*, 2025.
31. **Md Erfan**, A. Ryan, and M. R. Rahman. "Investigating Software Security Testing Challenges in Developer Discussions." Available at SSRN.

Teaching Experience

The University of Alabama

Spring 2026 **Teaching Assistant – Program Analysis and Software Interface Design**
- Supported classroom instruction and student learning.
- Assisted with assignments, grading, and course-related questions.
- Provided instructional support for program analysis and software design concepts.

Fall 2025 **Teaching Assistant – Database Systems**
- Assisted with database course instruction, assignments, grading, and student questions.
- Supported students with database concepts, SQL, and related coursework.

University of Barishal

2016-2025 • Taught undergraduate courses including Computer Algorithms, Data Structures, Mathematical Analysis for Computer Science, Numerical Methods, Peripheral Systems, Technical Writing, Data Communication, English Communication Skills, Software Engineering, and Information Systems Design.

• Taught graduate courses including Software Quality Assurance, Internet of Things (IoT), and Cloud Computing.

• Designed course materials, assignments, examinations, and laboratory exercises to support student learning.

• Supervised approximately 50 student research projects and provided hands-on guidance to undergraduate and graduate students.

Projects

2019–2021 **KachaBazar**, *Farmer-Friendly Platform*, Technologies: Java, JSP, MySQL, Hibernate
Role: Developer and Application Architect

2017–2025 **University Admission System**, *University of Barishal*, Technologies: PHP, CodeIgniter, MySQL
Role: Developer, Analyst, and Tester

Standardized Tests

IELTS Overall: 6.5; Speaking: 6.5; Listening: 5.5; Reading: 7.0; Writing: 6.0

GRE Quantitative Reasoning: 158; Analytical Writing: 3.0

Awards and Achievements

2021–2023 Regional Mentor, NASA Space Apps Challenge

2015 Pre-Graduation Merit Award, University of Dhaka, for Outstanding Academic Performance

2016–Present Coach and Manager, University of Barishal Football and Athletic Teams

2015 **3rd Position** in the 5000m and 10000m races at the Bangladesh Inter-University Athletic Competition

2014–2015 **2nd Position** in the 400m, 800m, and 1500m races at the University of Dhaka Intra-Hall Athletics Competition

Research Impact

Google Scholar 140 citations; h-index: 7; i10-index: 4